BIRZEIT UNIVERSITY

Department of Computer Science

COMP4381(Data Science)

Project Phase3 Report

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1. Data Ingestion & Scope

- The foundation of this dataset was built by integrating two primary raw sources:
- airports.csv: Contains geographical and infrastructural metadata (e.g., location, type, country).
- Airports_Punctuality-Data.csv: Contains daily operational metrics, including delay and punctuality statistics.
- Integration Strategy: The datasets were merged using the airport's municipality as the primary key. To maintain regional focus, the final corpus was strictly filtered to include only airports within France, Germany, Italy, Spain, the Netherlands.

2. Data Cleaning & Quality Assurance

To ensure the integrity of downstream modeling, several rigorous cleaning steps were applied to reduce noise and handle missing values:

- **Dimensionality Reduction:** Redundant and irrelevant columns were dropped to optimize memory and reduce computational overhead.
- **Missing Value Handling:**
    - Records lacking the critical `AverageDelay` metric were entirely dropped.
    - Remaining null values in numerical columns were imputed using the **median** of their respective distributions to ensure robustness against outliers.
- **Deduplication:** All identical, duplicate rows were eliminated.
- **Outlier Removal:** Extreme delay anomalies were identified and filtered out using the **Interquartile Range (IQR)** method to prevent statistical skew.

3. Feature Engineering

To enrich the dataset for predictive modeling and deeper analysis, several new variables were synthesized from the raw data.

Feature Name	Data Type	Description & Derivation
AverageDelay	Continuous	The arithmetic mean of a flight's schedule delay across both departure and arrival.
Is_Delayed	Binary	The primary target variable for classification. Flags a flight as delayed (**1**) if the AverageDelay exceeds 15 minutes, and on-time (**0**) otherwise.
Date Metrics	Temporal	Parsed components from the raw timestamp, explicitly extracting the month, day of the week, and a boolean weekend indicator.
Season	Categorical	Meteorological mapping of the flight month into Winter, Spring, Summer, or Fall.
Airport Size	Categorical	A consolidated grouping of raw airport types into four standardized tiers: *Large*, *Medium*, *Small*, or *Other*.

4. Final Output

Deliverable: The culmination of this pipeline is a high-quality, enriched dataset designed for advanced analytics and machine learning applications. The finalized records have been exported and are available in cleaned.